



A Project Report

“AUTOINTEL AI”

Intelligent Vehicle Diagnosis and Maintenance Assistant

carried out as part of the Tata Motors Summer Internship Submitted

by

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Submitted to:
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ACKNOWLEDGEMENT

The internship opportunity provided by **TATA Motors Ltd.** has been a highly valuable and enriching experience, offering significant learning and professional development. We consider ourselves fortunate to have been part of this internship program, and we are sincerely grateful for the opportunity to contribute and grow within such a prestigious organization.

We would like to express our deepest gratitude to our project guide for his support and encouragement throughout our internship. Despite his demanding responsibilities, he took the time to guide us, ensure we remained on the right path, and allowed us to carry out our project within the esteemed premises of Tata Motors. His leadership and mentorship have been truly inspiring. His coordination and efforts in arranging the necessary resources and creating a conducive environment for our learning are deeply appreciated.

This internship has been a significant milestone in our career journey. The skills, knowledge, and experiences gained will undoubtedly shape our future endeavors. We look forward to applying what we have learned in meaningful ways and hope to continue our association with Tata Motors and its esteemed professionals in the future.

Company Overview

Tata Motors Limited is a leading global automobile manufacturer and a flagship company of the **USD 100 billion Tata Group, founded by Jamshedji Tata in 1868**. Established in 1945, Tata Motors has grown into a USD 52.33 billion enterprise with a global footprint, operating in over 175 countries and employing more than **81,000 professionals worldwide**.

The company's extensive product portfolio includes passenger vehicles, sports utility vehicles, buses, trucks, and defence vehicles, with its marque widely recognized both on and off-road across international markets. As India's largest automobile company, Tata Motors has consistently demonstrated leadership in customer-centric innovation and technological advancement.

Tata Motors is committed to transforming the automotive landscape through cutting-edge technology, efficient design, and sustainable practices. The company's commercial vehicles are known for their power, performance, and cost-efficiency, while its new-generation passenger vehicles offer enhanced comfort, connectivity, and driving experience.

With advanced design and R&D centres located in India, the UK, Italy, and South Korea, Tata Motors continues to invest in future-ready solutions and tech-enabled products aimed at meeting the evolving needs of the modern consumer. The organization places strong emphasis on sustainability, corporate responsibility, and the philosophy of giving back to society, which remains deeply embedded in its core values.

Tata Motors' mission is clear and inspiring: ***"To be passionate in anticipating and providing the best vehicles and experiences that excite customers globally."*** This vision continues to drive the company's innovation, growth, and commitment to excellence across all its business operations.



Abstract

The rapid advancement of modern automobiles has introduced sophisticated electronic control systems and onboard diagnostics, making vehicle fault detection increasingly complex for ordinary users. Most existing diagnostic methods require technical expertise, specialized OBD-II scanning tools, or assistance from professional mechanics, leading to increased time, cost, and inconvenience. To address these challenges, this project presents **AutoIntel AI – Intelligent Vehicle Diagnosis and Maintenance Assistant**, an AI-powered web application designed to simplify the vehicle diagnosis process through natural language processing and intelligent information retrieval.

The system enables users to describe vehicle issues in simple, everyday language without requiring prior automotive knowledge. The entered complaint is analysed using Natural Language Processing (NLP) techniques to identify the most probable vehicle fault. Based on the identified fault, the application retrieves the corresponding OBD-II diagnostic code, explains the possible causes, recommends appropriate repair solutions, estimates repair cost and repair time, and provides preventive maintenance suggestions. The system also utilizes historical vehicle maintenance records, where available, to generate a more comprehensive and context-aware diagnosis.

The application is developed using **Python** and **Streamlit** for the user interface, with **Pandas** for data preprocessing, **SQL** for structured data management, and **Sentence Transformers** along with similarity-based retrieval techniques for complaint analysis. Multiple datasets, including Vehicle Master, OBD-II Knowledge Base, Repair Knowledge, and Vehicle Maintenance records, are integrated to provide accurate and meaningful diagnostic results.

The developed prototype demonstrates that artificial intelligence can significantly improve the accessibility and efficiency of vehicle diagnostics by reducing dependence on technical expertise and providing users with clear, actionable recommendations. This internship project enhanced practical knowledge in AI application development, data preprocessing, database integration, natural language processing, and software testing while showcasing the potential of intelligent diagnostic systems in the automotive domain.

Introduction

Modern vehicles are equipped with advanced electronic systems and onboard diagnostic (OBD-II) technology to monitor their performance and detect faults. While these advancements have improved vehicle reliability and safety, diagnosing faults has become increasingly complex. Most vehicle owners rely on professional mechanics or diagnostic scanners to identify problems, which can be time-consuming, expensive, and often requires technical expertise.

With the rapid growth of Artificial Intelligence (AI) and Natural Language Processing (NLP), intelligent systems can now interpret user queries and provide meaningful insights without requiring specialized knowledge. Applying these technologies to vehicle diagnostics can simplify the troubleshooting process and make maintenance recommendations more accessible to everyday users.

This internship project focuses on the development of **AutoIntel AI – Intelligent Vehicle Diagnosis and Maintenance Assistant**, an AI-powered application that bridges the gap between vehicle owners and technical diagnostic systems. The application enables users to describe vehicle problems in simple language and assists them in understanding potential faults through an intelligent, user-friendly interface. By combining automotive knowledge with AI-driven analysis, the system aims to improve the efficiency, accessibility, and reliability of preliminary vehicle diagnosis while promoting informed maintenance decisions.

Objectives of the Project

The primary objective of **Auto Intel AI** is to develop an AI-powered vehicle diagnostic system that simplifies fault identification and assists users in making informed maintenance decisions through intelligent analysis of vehicle complaints.

- 1. Develop an Intelligent Vehicle Diagnosis System-** To design and develop a web-based application capable of analysing vehicle complaints and providing accurate diagnostic information using Artificial Intelligence and Natural Language Processing (NLP).
- 2. Simplify Vehicle Fault Identification-** To enable users to describe vehicle problems in simple, everyday language, eliminating the need for technical knowledge or specialized diagnostic expertise.
- 3. Automate Complaint Analysis-** To process user complaints intelligently and identify the most probable vehicle fault by comparing the input with a structured automotive knowledge base.
- 4. Retrieve OBD-II Diagnostic Information-** To automatically retrieve the corresponding OBD-II diagnostic code, its description, and severity level, helping users better understand the identified issue.
- 5. Provide Repair Recommendations-** To generate detailed repair guidance by presenting possible causes of the fault, recommended repair procedures, estimated repair cost, and expected repair time.
- 6. Integrate Multiple Automotive Datasets-** To combine Vehicle Master, Repair Knowledge, OBD-II, and Vehicle Maintenance datasets into a unified system for generating accurate and comprehensive diagnostic reports.
- 7. Recommend Preventive Maintenance-** To suggest preventive maintenance measures that help users reduce the occurrence of recurring vehicle faults and improve overall vehicle reliability.
- 8. Enhance User Experience-** To develop an intuitive and interactive web interface that allows users to navigate the application easily and obtain diagnostic results quickly.
- 9. Demonstrate the Application of AI in Automotive Diagnostics-** To showcase the practical use of Artificial Intelligence, Natural Language Processing, and database integration in solving real-world vehicle diagnostic and maintenance challenges.

Methodology and System Architecture

6.1 Methodology

The development of **AutoIntel AI – Intelligent Vehicle Diagnosis and Maintenance Assistant** follows a structured methodology that integrates Natural Language Processing (NLP), machine learning techniques, and multiple automotive knowledge bases to provide an intelligent vehicle diagnosis system. The overall workflow consists of the following stages:

Step 1: Vehicle Selection

The user begins by selecting the vehicle details, including the brand, model, fuel type, engine type, and manufacturing year. These details are retrieved from the Vehicle Master dataset and help narrow down the diagnosis by providing vehicle-specific information.

Step 2: Complaint Submission

The user enters the vehicle problem in natural language through the application's interface. Instead of requiring technical terminology, the system accepts complaints written in simple everyday language, making the application easy to use for non-technical users.

Step 3: Complaint Analysis

The entered complaint is preprocessed and analyzed using Natural Language Processing techniques. The system compares the user's complaint with previously stored complaint patterns and automotive knowledge to determine the most relevant vehicle fault. This intelligent matching process improves the accuracy of diagnosis and minimizes manual interpretation.

Step 4: Fault Identification

Based on the complaint analysis, the system predicts the most probable vehicle fault. The identified fault acts as the key for retrieving additional diagnostic information from the integrated knowledge base.

Step 5: OBD-II Code Retrieval

Once the fault is identified, the corresponding OBD-II (On-Board Diagnostics) code is retrieved from the OBD Knowledge Base. Along with the code, the system displays its description, severity level, and technical information, helping users better understand the nature of the fault.

Step 6: Repair Knowledge Retrieval

The application accesses the Repair Knowledge dataset to retrieve detailed information related to the detected fault. This includes possible causes, recommended repair procedures, estimated repair cost, expected repair time, and preventive maintenance suggestions.

Step 7: Maintenance History Analysis

If historical service records are available for the selected vehicle, the system examines previous maintenance activities. This provides additional context for the diagnosis and assists users in understanding recurring issues or overdue maintenance.

Step 8: Final Diagnosis Report

Finally, all retrieved information is consolidated into a comprehensive diagnosis report. The report includes vehicle details, user complaint, detected fault, OBD-II code, possible causes, repair recommendations, estimated repair cost, estimated repair time, maintenance history, and preventive maintenance suggestions. This enables users to make informed decisions before consulting a service center.

6.2 System Architecture

The architecture of AutoIntel AI is designed as a modular system in which each component performs a specific function while working together to generate an intelligent diagnosis.

User Interface Layer

The front-end of the application is developed using **Streamlit**, providing an interactive interface where users can select vehicle details, enter complaints, and view the generated diagnosis report.

Complaint Analysis Engine

This module processes the user's complaint using Natural Language Processing techniques. It analyzes the text, extracts meaningful information, and compares it with the complaint knowledge base to identify the most relevant vehicle fault.

Knowledge Base Layer

The knowledge base serves as the core of the application and integrates multiple datasets:

- Vehicle Master Dataset
- OBD-II Knowledge Base
- Repair Knowledge Dataset
- Vehicle Maintenance Dataset

These datasets collectively provide vehicle information, diagnostic codes, repair recommendations, and maintenance history required for accurate diagnosis.

Database Layer

The processed datasets are stored in an SQL database, enabling efficient storage, retrieval, and management of automotive information. The database ensures fast querying and maintains data consistency across different application modules.

Diagnosis Engine

The diagnosis engine combines the outputs from the complaint analysis module with information retrieved from the knowledge base. It identifies the most probable fault and gathers all relevant diagnostic information required for report generation.

Report Generation Module

The final module compiles the diagnosis results into a structured report containing:

- Vehicle Information
- User Complaint
- Predicted Fault
- OBD-II Code
- Fault Description
- Possible Causes
- Recommended Repair Solution
- Estimated Repair Cost
- Estimated Repair Time
- Preventive Maintenance Recommendations

The generated report provides users with a comprehensive overview of the detected issue and serves as a preliminary diagnostic guide before professional inspection.

Implementation and Results

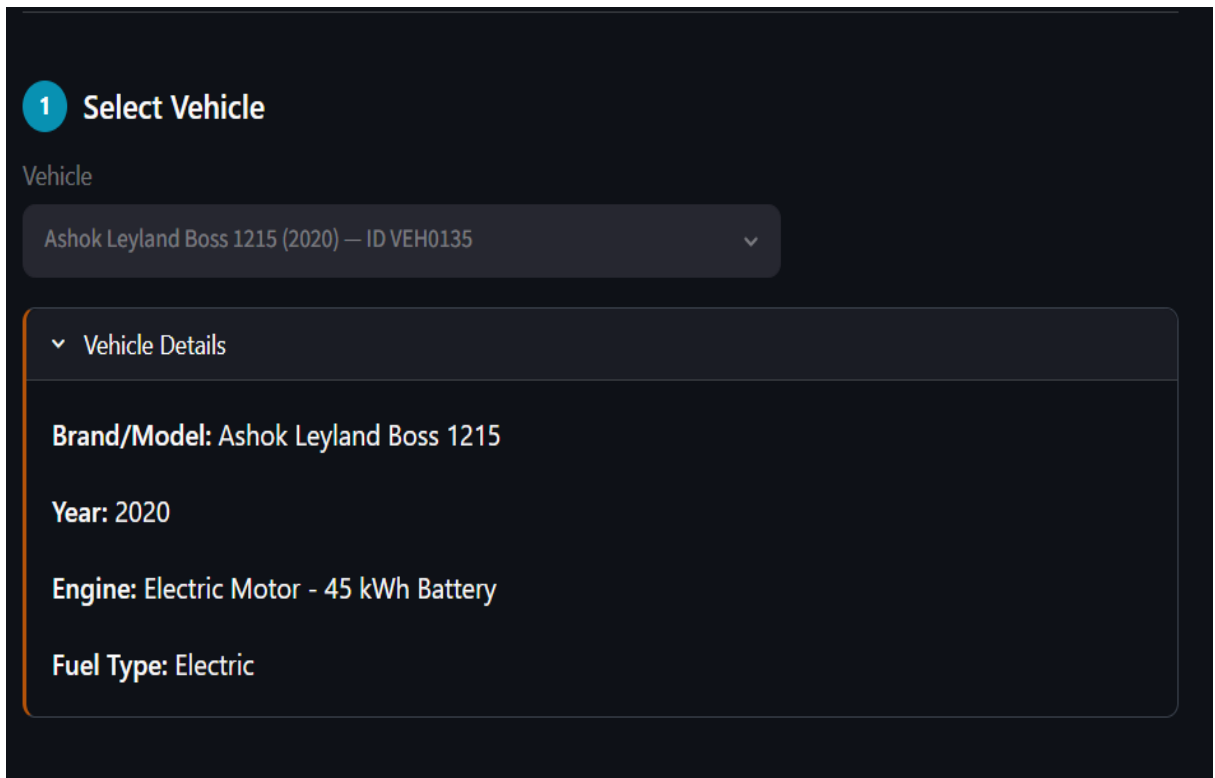
7.1 System Implementation

The AutoIntel AI application was developed as a web-based vehicle diagnostic assistant using Python and Streamlit. The system integrates multiple automotive datasets with Natural Language Processing (NLP) techniques to provide intelligent vehicle fault diagnosis. The implementation consists of several interconnected modules, each responsible for a specific stage of the diagnostic process.

7.1.1 Vehicle Selection Module

The first stage of the application allows the user to select a vehicle from the available Vehicle Master database. Once selected, the system displays essential vehicle details such as brand, model, manufacturing year, engine specifications, and fuel type. This information provides context for the diagnosis and ensures that subsequent recommendations are relevant to the selected vehicle.

Figure 7.1: Vehicle Selection Interface



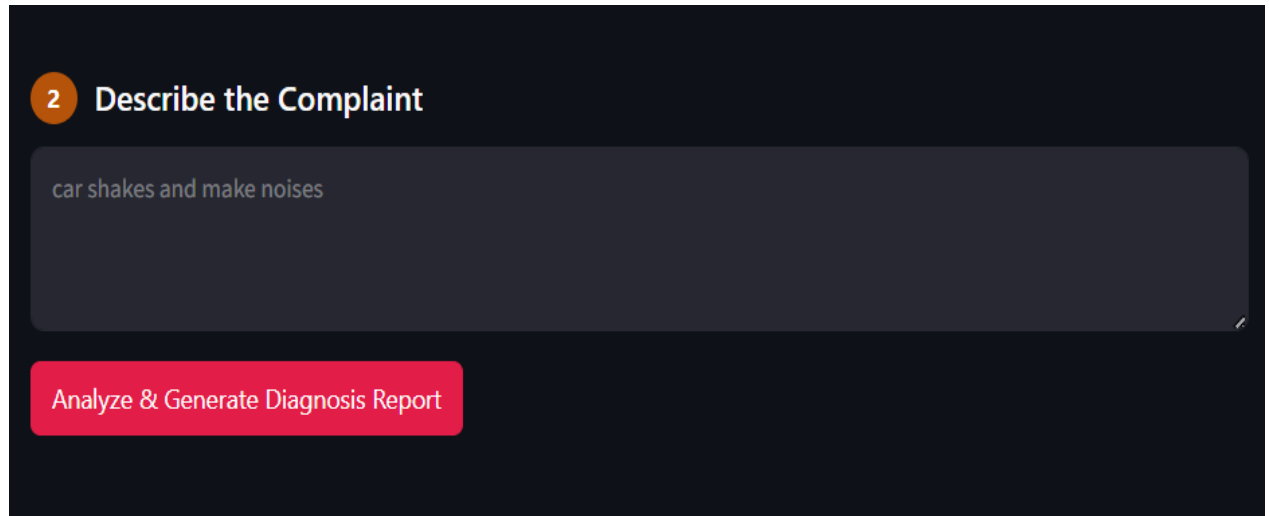
The screenshot displays a web interface for vehicle selection. At the top, a teal circle with the number '1' is followed by the heading 'Select Vehicle'. Below this, a label 'Vehicle' is positioned above a dark grey dropdown menu. The dropdown menu is open, showing the text 'Ashok Leyland Boss 1215 (2020) — ID VEH0135' with a downward arrow on the right. Below the dropdown, a section titled 'Vehicle Details' with a downward arrow is expanded, revealing four lines of text: 'Brand/Model: Ashok Leyland Boss 1215', 'Year: 2020', 'Engine: Electric Motor - 45 kWh Battery', and 'Fuel Type: Electric'.

7.1.2 Complaint Analysis Module

After selecting the vehicle, the user enters the observed vehicle problem in natural language. The application accepts complaints written in simple English without requiring technical automotive terminology. This makes the system accessible to both technical and non-technical users.

The complaint is then processed by the NLP engine, where keywords are extracted and compared against historical complaint records stored in the knowledge base.

Figure 7.2: Complaint Input Screen



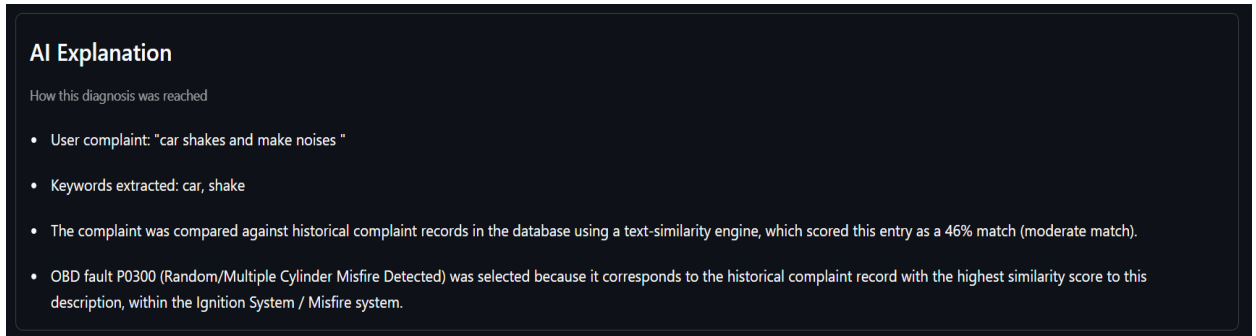
7.1.3 AI-Based Diagnosis Engine

The diagnosis engine performs similarity-based complaint matching to identify the most probable vehicle fault. The input complaint is compared with historical complaint records using text similarity techniques, and the fault with the highest similarity score is selected.

For the sample complaint *"car shakes and make noises"*, the system extracted the keywords **car** and **shake** and identified **OBD Code P0300 (Random/Multiple Cylinder Misfire Detected)** as the most probable diagnosis with a similarity score of **46%**, indicating a moderate match.

The system also displays the reasoning behind the diagnosis, improving transparency and helping users understand how the result was generated.

Figure 7.3: AI Explanation Module



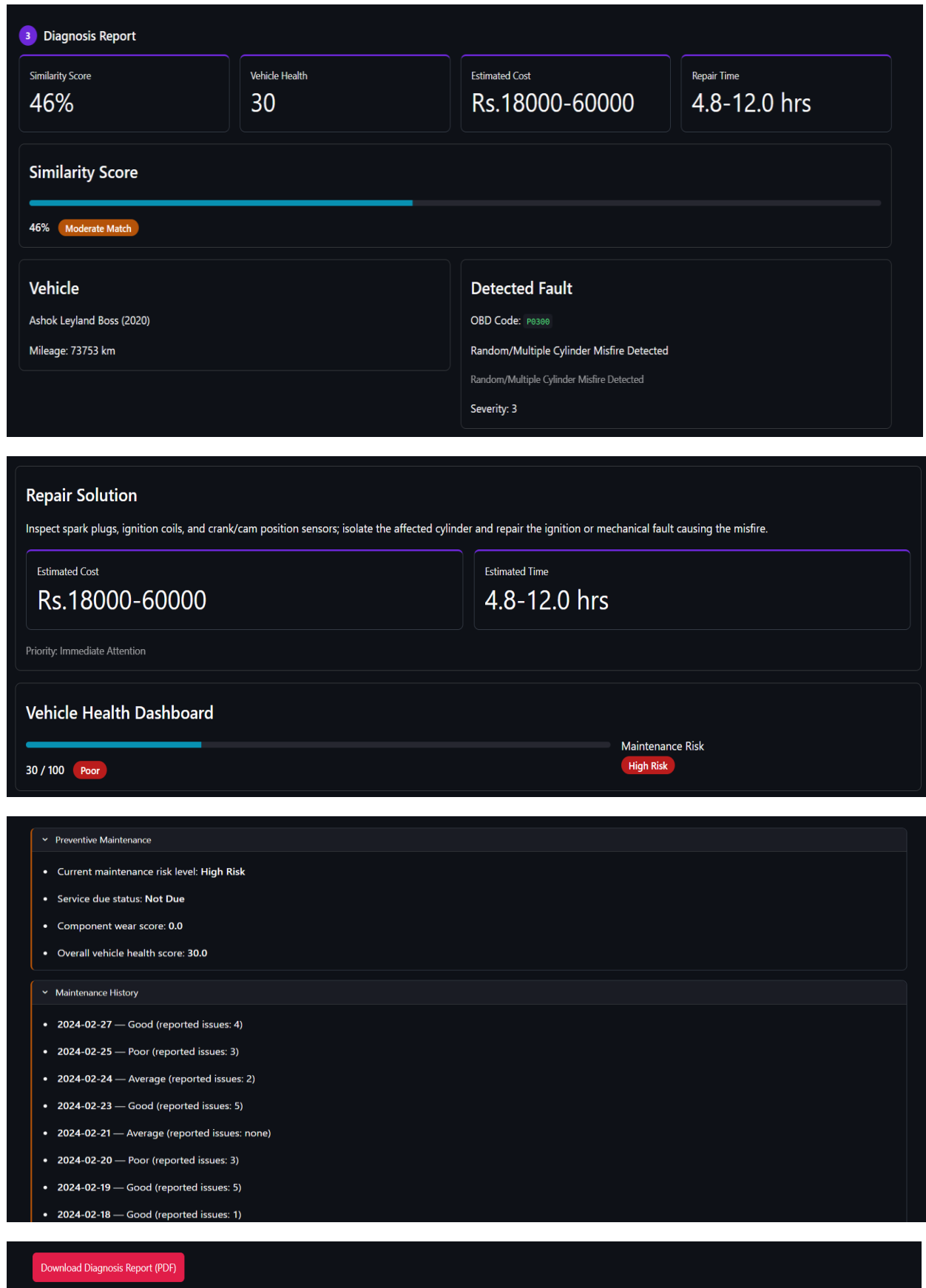
7.1.4 Diagnosis Report Generation

Once the diagnosis is completed, the application generates a comprehensive diagnostic report containing:

- Vehicle Information
- Similarity Score
- Detected Fault
- OBD-II Diagnostic Code
- Fault Severity
- Repair Solution
- Estimated Repair Cost
- Estimated Repair Time
- Vehicle Health Score
- Maintenance Risk Assessment
- Preventive Maintenance Recommendations
- Previous Maintenance History

The report can also be downloaded as a PDF for future reference.

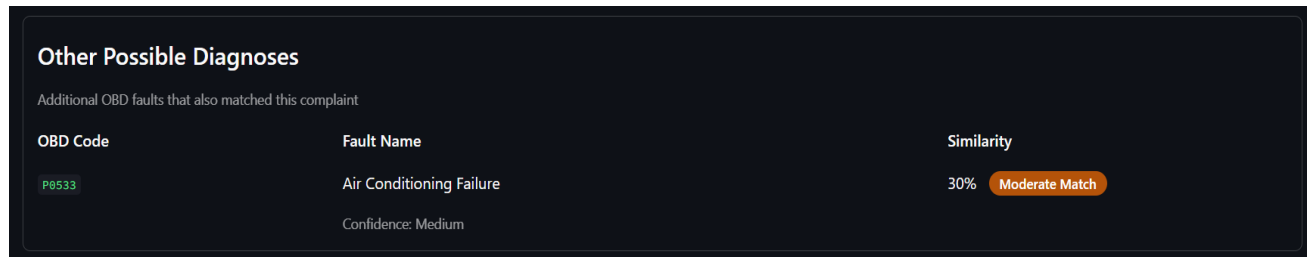
Figure 7.4: Generated Diagnosis Report



7.1.5 Alternative Diagnosis Recommendation

Since vehicle complaints may correspond to multiple faults, the application also displays other possible diagnoses ranked according to their similarity scores. This feature assists users and technicians by presenting alternative faults that should also be considered during inspection.

Figure 7.5: Other Possible Diagnoses



OBD Code	Fault Name	Similarity
P0533	Air Conditioning Failure	30% Moderate Match

Confidence: Medium

7.2 Results

The developed AutoIntel AI system successfully demonstrates the practical application of Artificial Intelligence in preliminary vehicle diagnostics. The application accepts user complaints in natural language, analyses them using similarity-based matching techniques, retrieves relevant OBD-II diagnostic information, and generates a comprehensive diagnosis report.

The experimental results indicate that the system is capable of:

- Identifying probable vehicle faults based on user complaints.
- Retrieving corresponding OBD-II diagnostic codes and severity levels.
- Providing repair recommendations, estimated repair cost, and expected repair time.
- Displaying preventive maintenance suggestions and historical maintenance records.
- Offering explainable AI output by showing extracted keywords, similarity scores, and diagnosis reasoning.
- Suggesting multiple possible diagnoses when similar complaints exist, thereby improving decision support.

The web-based interface provides an intuitive user experience by presenting all diagnostic information in an organized dashboard. Features such as vehicle health assessment, maintenance risk visualization, downloadable PDF reports, and explainable AI improve both usability and transparency. Overall, the developed prototype demonstrates that integrating Natural Language Processing with automotive knowledge bases can significantly simplify vehicle fault diagnosis and provide users with reliable preliminary maintenance guidance.

Technology Gaps & Future Work

As part of our digital defect tracking platform, we identified several areas with potential for technical enhancement. While the current system serves as a functional prototype, the following aspects highlight existing limitations and our roadmap for future improvements

1. Complaint Analysis Engine

- **Current Status:**

The application currently uses Natural Language Processing (NLP) with similarity-based text matching to analyze user complaints and identify the most relevant vehicle fault. The system also provides an AI explanation showing extracted keywords and similarity scores.

- **Future Work:**

The complaint analysis engine can be enhanced using transformer-based Large Language Models (LLMs) to better understand complex, ambiguous, and context-rich vehicle complaints, thereby improving diagnostic accuracy.

2. Vehicle Data Integration

- **Current Status:**

The platform integrates Vehicle Master, OBD-II, Repair Knowledge, and Vehicle Maintenance datasets to generate comprehensive diagnostic reports.

- **Future Work:**

The knowledge base can be expanded to include a wider range of vehicle manufacturers, models, variants, and additional fault records to improve coverage and diagnostic reliability.

3. Vehicle Diagnosis

- **Current Status:**

The system predicts the most probable vehicle fault based on user-entered complaints and displays alternative diagnoses with similarity scores for better decision support.

- **Future Work:**

Future versions can support multiple complaint analysis, allowing the system to identify multiple related faults simultaneously and generate a consolidated diagnosis report.

4. Data Storage

- **Current Status:**

The application currently uses an SQLite database to store vehicle information, OBD-II codes, repair knowledge, and maintenance records, providing a lightweight and efficient storage solution for the prototype.

- **Future Work:**
The system can be migrated to cloud-based databases such as PostgreSQL or MongoDB to improve scalability, concurrent user support, data security, and real-time synchronization.

5. Vehicle Diagnostics

- **Current Status:**
The diagnosis is generated using user-entered complaints and historical automotive datasets without requiring direct communication with the vehicle.
- **Future Work:**
The application can be integrated with real-time OBD-II scanners to automatically retrieve live diagnostic trouble codes and sensor data, enabling more accurate and reliable fault detection.

6. Repair Cost Estimation

- **Current Status:**
Repair cost and repair time estimates are retrieved from the repair knowledge database and presented as approximate ranges.
- **Future Work:**
Future versions can integrate service center and spare parts databases to provide location-specific, real-time repair cost estimates.

7. User Interface

- **Current Status:**
The system provides an interactive Streamlit-based dashboard featuring vehicle selection, AI explanation, vehicle health visualization, diagnosis reports, and downloadable PDF reports.
- **Future Work:**
The interface can be extended into a dedicated Android and iOS application with voice-based complaint input and multilingual support to improve accessibility and user experience.

8. Predictive Maintenance

- **Current Status:**
The application provides preventive maintenance recommendations based on predefined repair knowledge and historical maintenance records.
- **Future Work:**
Machine learning models can be incorporated to predict future component failures using historical maintenance data and vehicle usage patterns, enabling predictive maintenance instead of reactive repairs.

Conclusion

The **AutoIntel AI – Intelligent Vehicle Diagnosis and Maintenance Assistant** project successfully demonstrates the application of Artificial Intelligence and Natural Language Processing in simplifying preliminary vehicle fault diagnosis. By allowing users to describe vehicle issues in natural language, the system intelligently identifies the most probable fault, retrieves the corresponding OBD-II diagnostic code, and provides detailed repair recommendations, estimated repair cost, repair time, preventive maintenance suggestions, and vehicle health insights through an intuitive web interface.

The integration of multiple automotive datasets, similarity-based complaint analysis, explainable AI, and automated report generation provides users with a comprehensive and accessible diagnostic experience. Features such as alternative fault suggestions, maintenance history, and downloadable diagnosis reports further enhance the usability and practical value of the application.

This internship project provided valuable hands-on experience in AI application development, Natural Language Processing, database management, software integration, and web application development using Python and Streamlit. It also strengthened problem-solving, testing, and system design skills while demonstrating how intelligent technologies can be applied to solve real-world challenges in the automotive domain.

Overall, AutoIntel AI serves as a promising prototype for intelligent vehicle diagnostics and establishes a strong foundation for future enhancements such as real-time OBD-II integration, cloud-based deployment, advanced AI models, and predictive maintenance capabilities.

Key Takeaways

The internship at Tata Motors Ltd., combined with the The internship project "AutoIntel AI – Intelligent Vehicle Diagnosis and Maintenance Assistant" provided valuable exposure to Artificial Intelligence, Natural Language Processing, and intelligent software development. Working on this project enabled the practical application of theoretical concepts while developing a real-world solution for preliminary vehicle diagnostics. The following are the major takeaways from this experience:

AI-Based Problem Solving

- Gained hands-on experience in developing an intelligent vehicle diagnosis system capable of analyzing user complaints written in natural language.
- Learned how Artificial Intelligence and Natural Language Processing can simplify complex automotive diagnostic processes and improve user accessibility.

Natural Language Processing and Information Retrieval

- Developed an understanding of text preprocessing, keyword extraction, similarity-based complaint matching, and intelligent information retrieval techniques.
- Learned how user complaints can be analyzed to identify the most probable vehicle fault and generate meaningful diagnostic recommendations.

Database Integration and Knowledge Management

- Worked with multiple automotive datasets, including Vehicle Master, OBD-II, Repair Knowledge, and Vehicle Maintenance records.
- Understood the importance of structured data management and database integration in building reliable AI-powered applications.

Web Application Development

- Designed and developed an interactive web application using Python and Streamlit to provide an intuitive interface for vehicle selection, complaint analysis, and diagnosis report generation.
- Strengthened knowledge of frontend interaction, backend logic, report generation, and user-centered application design.

Explainable AI and Decision Support

- Learned the importance of transparency in AI systems by implementing features such as similarity scores, extracted keywords, AI-generated explanations, and alternative diagnosis recommendations.
- Gained insights into developing AI solutions that assist users in understanding how diagnostic decisions are made.

System Integration and Testing

- Acquired practical experience in integrating multiple software modules, databases, and knowledge bases into a unified diagnostic system.
- Performed testing and validation to ensure accurate diagnosis, smooth workflow, and reliable system performance.

Professional and Technical Development

- Enhanced problem-solving, analytical thinking, software development, and technical documentation skills through the successful implementation of an end-to-end AI application.
- Improved teamwork, project planning, debugging, and communication skills while working collaboratively throughout the project lifecycle.

Career-Oriented Learning

- The project provided valuable practical experience in applying Artificial Intelligence, Natural Language Processing, and database technologies to solve real-world automotive problems.
- Strengthened readiness for future opportunities in Artificial Intelligence, Data Science, Software Development, Intelligent Systems, and Digital Transformation.

Overall, the internship provided an excellent opportunity to apply academic concepts to a practical automotive application while strengthening technical expertise in Artificial Intelligence, Natural Language Processing, database management, and software development. The experience has laid a strong foundation for future work in intelligent systems and AI-driven automotive solutions.